MoleculeLens: Closed-Form Linear Saliency for Drug–Text Retrieval under Scaffold Shift

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Abstract

We present *MoleculeLens*, a framework for interpretable drug–text retrieval that makes every alignment decision traceable to molecular substructures. The model combines frozen ECFP4 fingerprints and a frozen biomedical sentence encoder with learned linear projection heads in a shared embedding space. Because the molecule-side map is linear, the model admits a closed-form linear saliency approximation over ECFP4 bits; on held-out scaffold pairs, this proxy tracks the exact normalized-score gradient strongly (mean Pearson 0.905) and the exact leave-one-bit-out score even more closely (mean Pearson 0.947). We evaluate the framework on a reproducible ChEMBL benchmark of 2,699 approved-drug mechanism pairs under a Bemis–Murcko scaffold-disjoint split designed to reduce memorization and expose lexical leakage. On this harder setting, MoleculeLens remains competitive with retrieval baselines while enabling analyses that black-box models cannot: pharmacophore-level explanations for correct matches, attribution fidelity diagnostics for failed retrievals, a same-model substructure-level account of drug-name leakage, and a descriptive layer-wise view of where drug-retrievable information emerges in the frozen text encoder. These results position MoleculeLens as an auditable approach to studying what molecular features drive drug–text alignment under realistic scaffold shift.

1 Introduction

Drug–text alignment asks whether molecular structures and natural-language descriptions of mechanism can be embedded in a common space for retrieval. A robust solution would support mechanism-based search, structured pharmacology retrieval, and retrieval-augmented drug discovery workflows. In practice, however, current evaluations leave two questions unresolved. First, strong retrieval scores can arise from shortcut signals such as drug names, target metadata, or scaffold-correlated memorisation rather than genuine structure–mechanism correspondence. Second, even when retrieval succeeds, standard dual-encoder models provide little evidence about *which molecular features actually made the match possible*.

MoleculeLens is designed around these two problems. It uses fixed ECFP4 fingerprints for molecules and a frozen biomedical text encoder for mechanism descriptions, and learns only linear projections into a shared embedding space. This deliberately constrained architecture is not merely a simplification choice: it makes each retrieval decision analyzable with a fast closed-form saliency proxy. By projecting a text embedding back through the molecule-side map, we obtain a linear saliency score for each ECFP4 bit and decode the top-scoring bits to concrete molecular substructures. No post-hoc explainer or sampling-based approximation is required, and in Section 7 we validate the proxy against exact gradients and exact leave-one-bit-out score drops.

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To study this in a setting where retrieval quality is harder to inflate, we construct a reproducible ChEMBL benchmark of 2,699 approved-drug mechanism pairs under a Bemis–Murcko scaffold-disjoint split. The benchmark reduces opportunities to memorise training scaffolds while preserving a chemically meaningful retrieval task. It also supports an explicit no-drug-name ablation, allowing us to separate mechanism grounding from lexical identity cues.

Within this setting, MoleculeLens serves two purposes. As a retrieval model, it provides a lightweight and interpretable baseline that remains competitive on held-out scaffolds. As an analysis tool, it turns the shared embedding space into something auditable: we can inspect the substructures behind correct matches, validate the linear saliency approximation against the exact normalized score, measure how attribution patterns shift when drug names are removed, and trace where drug-retrievable information appears across layers of the frozen text encoder.

This paper makes four concrete contributions:

1. **Validated closed-form linear saliency for drug–text retrieval.** We introduce an interpretable dual-encoder retrieval framework whose linear molecule-side projection enables closed-form saliency scores over ECFP4 bits, and we validate this proxy against exact gradients and exact leave-one-bit-out score changes.
2. **A reproducible scaffold-disjoint benchmark.** We construct a Bemis–Murcko scaffold-disjoint ChEMBL benchmark from 2,699 approved-drug mechanism pairs and evaluate MoleculeLens against MolPrompt, KV-PLM, and a zero-shot Graphormer baseline.
3. **A structural account of lexical leakage.** We show with a same-model text ablation that removing drug names changes not only retrieval accuracy but also the substructures that the fixed rich model treats as important, yielding a mechanistic view of leakage beyond aggregate score drops.
4. **A descriptive layer-wise diagnostic of text-side retrieval signal.** We adapt the logit-lens idea to contrastive retrieval and use it as a sanity check for where the frozen biomedical text encoder exposes drug-retrievable CLS information.

2 Related Work

Molecule–text representation learning. Recent work has explored joint learning over molecular structures and natural language, including retrieval-oriented models such as Text2Mol [Edwards et al., 2021], molecular language models such as MolT5 [Edwards et al., 2022] and BioT5 [Pei et al., 2023], graph-language models such as MolCA [Liu et al., 2023b], multimodal pretrained models such as KV-PLM [Zeng et al., 2022], and broader molecular benchmark suites such as MoleculeNet [Wu et al., 2018]. These systems differ in molecular modality, text domain, and supervision strategy, but they share a common assumption that chemically meaningful structure–text correspondences can be captured in a shared space. Our benchmark is complementary to these models: it isolates FDA-approved drug mechanism retrieval under scaffold shift and emphasizes auditable substructure-level explanations rather than maximum-capacity generation or captioning.

Graph and language baselines. Transformer architectures for graphs have become standard baselines for molecular prediction, with Graphormer as a representative example [Ying et al., 2021]. More recent 3D molecular pretraining systems such as Uni-Mol [Zhou et al., 2023] are important context for molecular representation learning, although they are not direct text-query retrieval baselines without an additional text-alignment head. In parallel, text encoders pretrained on scientific or biomedical corpora such as SciBERT [Beltagy et al., 2019] have influenced biomedical retrieval systems and text-conditioned molecular models. Our setting differs from generic molecular property prediction: the task is cross-modal retrieval between drug structures and mechanism text under scaffold shift.

Evaluation leakage and scaffold splits. Scaffold-based evaluation remains a standard strategy for reducing overoptimistic estimates in molecular machine learning [Bemis and Murcko, 1996]. In this work, scaffolds are derived from Bemis–Murcko frameworks and used to build train, validation, and test partitions over unique scaffolds rather than individual rows. The molecular side uses ECFP4 fingerprints [Rogers and Hahn, 2010] as a simple, strong structural representation. We also explicitly

test the effect of removing drug names from the text because identity-bearing lexical fields can inflate retrieval scores without improving true structure–mechanism grounding.

Mechanistic interpretability of representation spaces. Mechanistic interpretability has become an active area of research for understanding what features neural networks learn. Bricken et al. [Bricken et al., 2023] showed that sparse autoencoders (SAEs) can recover monosemantic features from superimposed polysemantic representations in transformer residual streams. Templeton et al. [Templeton et al., 2024] scaled this approach to frontier language models, identifying millions of interpretable features in Claude 3 Sonnet. Lindsey et al. [Lindsey et al., 2025] extended this to attribution graphs, providing causal accounts of specific model behaviors by tracing feature-to-feature influence. Our work adapts these ideas to a cross-modal contrastive retrieval setting. Rather than analysing a transformer residual stream, we analyse the 256-dimensional shared embedding space produced by two linear projection heads, and rather than token-level features, we decode back to ECFP4 molecular substructures [Rogers and Hahn, 2010] that have ground-truth chemical semantics via RDKit [RDKit contributors, 2023].

3 Benchmark and Task

3.1 ChEMBL Drug–Mechanism Pairs

The benchmark is constructed from ChEMBL [Zdrazil et al., 2024]. Starting from the mechanism endpoint, we keep approved drugs only by filtering for `max_phase=4` and molecular mechanisms with valid mechanism text. Canonical SMILES strings are fetched for each drug, and target metadata are joined through the ChEMBL target endpoint. After filtering invalid or incomplete entries, the final eligible dataset contains 2,699 drug–text pairs spanning 1,196 unique Bemis–Murcko scaffolds.

Each example contains a drug structure and a text description. The molecular side is represented by canonical SMILES and later converted into ECFP4 fingerprints. The text side is a concatenation of mechanism of action, target name, action type, and drug name:

$$\texttt{text_rich} = \texttt{mechanism} + \texttt{target} + \texttt{action} + \texttt{drug}. \tag{1}$$

This formulation is intentionally rich because the benchmark is designed both to measure retrieval accuracy and to expose sensitivity to lexical cues such as explicit drug-name mentions.

3.2 Retrieval Setup

The task is bidirectional retrieval: given a text query, retrieve the matching drug structure from a gallery, and given a structure query, retrieve the matching text. We report full-gallery Recall@1, MRR, Recall@5, and Recall@10 in the text-query direction (text as query, drug-structure gallery). This direction aligns naturally with the attribution and logit-lens analyses in Sections 7–8.

Evaluation protocol. All primary retrieval comparisons in this paper use single-label (diagonal) evaluation: a retrieval is correct if and only if the exact matching row is returned at rank 1. This ensures a consistent, apples-to-apples comparison across all models, since per-example retrieval scores are not available for MolPrompt and KV-PLM and therefore multi-label metrics cannot be computed for them.

Multi-label note. Several drugs appear under multiple mechanism rows because ChEMBL records each target–action pair separately. Diagonal evaluation is therefore slightly pessimistic: retrieving a different mechanism row of the same physical drug is not a model failure. Of 435 test rows, 193 belong to 76 parent structures with multiple mechanism rows after RDKit `FragmentParent` standardisation. The attribution analyses in Section 7 account for this by identifying same-parent-structure pairs from standardized SMILES and using parent-normalized identifiers where appropriate.

3.3 Scaffold Split

The main evaluation uses a saved scaffold manifest built over unique Bemis–Murcko scaffolds with seed 0. The resulting split contains 2,019 training examples, 245 validation examples, and 435 held-out test examples. The key design choice is that the test molecules belong to scaffolds not seen

during training, which makes the benchmark substantially harder than the earlier global split and better aligned with the intended generalization claim.

4 Method

4.1 Linear Contrastive Bridge

MoleculeLens uses a deliberately simple dual-encoder design. Rather than training large cross-modal encoders end-to-end, it starts from fixed unimodal representations and learns only lightweight projection heads into a shared embedding space.

For molecules, we compute ECFP4 fingerprints with radius 2 and 2,048 bits [Rogers and Hahn, 2010]. For text, we encode `text_rich` with a frozen biomedical sentence encoder and use the CLS-style pooled representation. Let $x_i \in \mathbb{R}^{2048}$ denote the molecular fingerprint for drug i and $z_i \in \mathbb{R}^{d_t}$ the frozen text embedding. MoleculeLens learns linear projections

$$b_i^{(m)} = W_m x_i, \qquad b_i^{(t)} = W_t z_i, \tag{2}$$

where both projected embeddings live in a 256-dimensional shared space and are L_2 normalized before scoring.

4.2 Training Objective

Training uses a symmetric contrastive loss over matched drug–text pairs. Given a minibatch of projected text embeddings $B^{(t)}$ and molecular embeddings $B^{(m)}$, similarities are computed with a temperature-scaled cosine form:

$$s_{ij} = \frac{\langle \hat{b}_i^{(t)}, \hat{b}_j^{(m)} \rangle}{\tau}. \tag{3}$$

The loss combines bidirectional InfoNCE with two modifications used in the current codebase: same-target weighting for hard negatives that share a target label, and a margin term against the hardest same-target negative in the minibatch. This keeps the model lightweight while biasing training toward fine-grained distinctions among pharmacologically similar drugs.

5 Experimental Setup

5.1 Baselines

We compare MoleculeLens against four families of baselines already present in the experimental repository.

- **MoleculeSTM (zero-shot):** an earlier molecular structure–text model [Liu et al., 2023a] evaluated from prior head-to-head runs.
- **Graphormer (zero-shot):** a graph-based baseline evaluated on the aligned scaffold split.
- **MolPrompt:** a trained retrieval baseline on the same held-out ChEMBL-style split.
- **KV-PLM:** a biomedical language-model baseline that encodes both structure strings and text in a shared representation [Zeng et al., 2022].

We also retain the earlier **Thin Bridges (Global)** result as a reference point. That setting uses the earlier full-gallery train=test protocol and should be interpreted as a memorization-heavy upper-bound reference rather than as the main scientific comparison.

5.2 Implementation Notes

The current scaffold model uses batch size 512, learning rate 10^{-3}, weight decay 10^{-4}, temperature 0.07, margin 0.1, and 100 training epochs. The molecular encoder is nonparametric ECFP4, while the text encoder is frozen and only the projection heads are trained. We retain a separately retrained no-drug condition for the baseline comparison tables, but the mechanistic leakage analysis in Section 7.4

Table 1: Primary retrieval comparison. All models use single-label (diagonal) evaluation for consistent apples-to-apples comparison. Full-gallery rows are included for context only; the held-out scaffold rows are the main scientific comparison.

Model	Eval Set	Recall@1	MRR	Recall@5	Recall@10
MoleculeSTM (zero-shot)	Full gallery (N=2699)	0.089	0.171	0.243	0.337
Thin Bridges (Global)	Full gallery (N=2699)	0.747	0.849	0.985	0.997
MolPrompt (Global)	Full gallery (N=2699)	0.123	0.264	0.405	0.590
KV-PLM (Global)	Full gallery (N=2699)	0.015	0.038	0.044	0.074
Graphormer (zero-shot)	Scaffold test (N=435)	0.002	0.015	0.011	0.025
MoleculeLens (ours)	Scaffold test (N=435)	**0.225**	**0.304**	0.384	0.462
MolPrompt	Scaffold test (N=435)	0.129	0.256	**0.400**	**0.531**
KV-PLM	Scaffold test (N=435)	0.055	0.121	0.172	0.251

Table 2: MoleculeLens text-field ablation on the canonical scaffold split. The mechanism-only row removes target name, action type, and explicit drug name.

Text condition	Recall@1	MRR	Recall@5	Recall@10
`text_rich`	0.225	0.304	0.384	0.462
`mechanism_of_action` only	0.124	0.226	0.331	0.453

uses a stricter *same-model* ablation: the rich model is held fixed and only the `Drug: X.` span is removed at inference time. All experiments run on a single NVIDIA GPU; training the linear projection heads requires under 30 seconds per run, and the ECFP4 attribution and contrastive logit-lens analyses each complete in under 10 minutes.

6 Results

6.1 Primary Retrieval Comparison

Table 1 shows the main comparison. MoleculeLens achieves the strongest Recall@1 and MRR among the held-out scaffold models, improving on MolPrompt by roughly ten Recall@1 points and on KV-PLM by seventeen Recall@1 points (all models: single-label diagonal evaluation). The zero-shot Graphormer baseline performs near chance on this retrieval task. For additional context, we also include available full-gallery reference runs for MolPrompt and KV-PLM. These results suggest that the thin-bridge design remains competitive when evaluated on unseen scaffolds rather than on the earlier full-gallery setup.

Two points matter for interpretation. First, the global Thin Bridges result remains much higher than every other full-gallery or scaffold-disjoint result, which reinforces that the earlier setting should be viewed as a reference rather than a fair main comparison. Second, MolPrompt attains stronger Recall@10 than MoleculeLens on the scaffold split, suggesting that MoleculeLens is better at placing the correct match near the top of the list but may still lose ground on broader candidate coverage.

Because the task is cross-modal, Table 1 is restricted to methods that score text queries against molecular representations. Same-modality text lookup controls are not included as competing baselines because they access gallery text rather than molecular structure. Instead, we audit lexical shortcutting through the no-drug-name ablation and the stricter same-model perturbation analysis in Section 7.4.

To directly test whether the target and action fields create a partial target-identity lookup signal, we additionally train MoleculeLens on `mechanism_of_action` alone, removing the target name, action type, and explicit drug field. This mechanism-only ablation reaches Recall@1 $= 0.124$ and MRR $= 0.226$ on the same 435-pair scaffold test set (Table 2). The drop from the rich-text model (0.225 to 0.124) confirms that target/action metadata contributes useful signal, but the mechanism-only result remains well above chance and close to the MolPrompt scaffold Recall@1 of 0.129.

Bootstrap 95% confidence intervals for the scaffold test set (N=435) are reported in Table 3. For MoleculeLens the CI is computed by exact bootstrap resampling (10,000 iterations) of per-example

Table 3: Bootstrap 95% confidence intervals for Recall@1 on the scaffold test set (N=435). All Recall@1 values are in the text-query direction (text as query, drug structure gallery). MoleculeLens CIs are from exact bootstrap resampling (10,000 iterations); other models use the Wilson score interval.

Model	Recall@1	95% CI (lo)	95% CI (hi)
MoleculeLens (text_rich)	0.225	0.186	0.267
MoleculeLens (text_nodrug)	0.110	0.083	0.140
MoleculeLens (text_nodrug, same-model)	0.067	0.044	0.092
MolPrompt	0.129	0.101	0.164
KV-PLM	0.055	0.037	0.081
Graphormer (zero-shot)	0.002	0.000	0.012

Table 4: Multi-seed scaffold split variance for MoleculeLens (text_rich and text_nodrug). Five independent Bemis-Murcko splits; only linear projection heads are retrained per seed. Leakage drop (%) $= (R@1_{\text{rich}} - R@1_{\text{nodrug}})/R@1_{\text{rich}} \times 100$.

Seed	n_{test}	R@1 (rich)	R@5 (rich)	R@10 (rich)	MRR (rich)	Leakage drop (%)
0	403	0.191	0.377	0.459	0.287	29.9
1	261	0.226	0.479	0.544	0.342	20.3
2	224	0.188	0.393	0.496	0.296	2.4
3	221	0.285	0.525	0.634	0.402	27.0
4	395	0.175	0.352	0.446	0.269	34.8
Mean	–	**0.213**	**0.425**	**0.516**	**0.319**	**22.9**
Std	–	0.045	0.070	0.073	0.053	12.6

retrieval scores. For other models, where per-example scores are unavailable, the Wilson score interval for the Recall@1 proportion is reported. The non-overlapping CIs between MoleculeLens (text_rich) $[0.186, 0.267]$ and MolPrompt $[0.101, 0.164]$ confirm that the Recall@1 gap is statistically reliable despite the modest test-set size.

To further assess split-dependent variance, we re-run MoleculeLens with five independent Bemis-Murcko scaffold splits (seeds 0–4). All 2,699 drug texts are encoded once; only the linear projection heads (W_m, W_t) are retrained per seed ($\approx$22 s each). Note that these multi-seed runs use a simplified InfoNCE objective (no same-target weighting or margin term) and a 90/10 train–test partition without a held-out validation set; they are therefore best interpreted as a *partition-sensitivity analysis* with a comparable proxy model rather than as a direct replication of the main scaffold training protocol. Table 4 reports per-seed results along with the mean and standard deviation. Recall@1 varies between 0.175 and 0.285 (mean 0.213 ± 0.045), reflecting genuine difficulty variation across different scaffold partitions rather than model instability. The drug-name leakage drop ranges from 2.4% (seed 2) to 34.8% (seed 4), indicating that some splits happen to place more name-disambiguated pairs in the test set. Figure 2 summarises all three robustness analyses in a single panel.

Because the five-seed table is intentionally a fast proxy, we also run a five-seed direct check with the full MoleculeLens loss: same-target weighting plus the hardest-same-target margin term used by the main model. These were the same consecutive seeds 0–4 used in the proxy analysis and were selected before evaluating the full-loss results. This full-loss check uses the same cached frozen embeddings and independent 90/10 scaffold splits. Table 5 shows that the complete-loss model is not less stable than the proxy: R@1 ranges from 0.203 to 0.407, with mean 0.277 ± 0.081. The result does not replace the canonical seed-0 scaffold split, but it directly addresses whether the simplified proxy was artificially inflating robustness; if anything, the full-loss check is stronger.

6.2 Drug-Name Leakage

Table 6 reports the *retrained* no-drug comparison used for apples-to-apples model benchmarking: each method is evaluated after removing explicit drug names from the text and recomputing its no-name retrieval outputs. As expected, all text-aware models degrade to some extent. The important comparison is between the old global Thin Bridges setup and the new scaffold model. The global Thin Bridges setting drops by 80.5% relative Recall@1, while global KV-PLM drops by 65.0% and

Table 5: Five-seed full-loss check. Unlike Table 4, these runs use the same same-target weighting and hardest-same-target margin term as the main MoleculeLens model. Seeds 0–4 were selected as the same consecutive seed set used by the proxy analysis before evaluating full-loss results.

Seed	n_{test}	R@1	R@5	R@10	MRR
0	403	0.221	0.335	0.424	0.291
1	261	0.295	0.479	0.533	0.381
2	224	0.259	0.375	0.442	0.328
3	221	0.407	0.561	0.629	0.484
4	395	0.203	0.342	0.451	0.284
Mean	–	**0.277**	**0.418**	**0.496**	**0.354**
Std	–	0.081	0.098	0.085	0.083

Table 6: Drug-name leakage ablation measured with Recall@1. All rows use retrained no-drug outputs for comparability across models; Section 7.4 reports the stricter same-model ablation for MoleculeLens only.

Model	Eval Set	With Drug	No Drug	Abs. Drop	% Drop
MoleculeSTM (zero-shot)	Full gallery (N=2699)	0.089	0.014	0.074	83.8
Thin Bridges (Global)	Full gallery (N=2699)	0.747	0.145	0.601	80.5
KV-PLM (Global)	Full gallery (N=2699)	0.015	0.005	0.010	65.0
Graphormer (zero-shot)	Scaffold test (N=435)	0.002	0.005	-0.002	—
MoleculeLens (ours)	Scaffold test (N=435)	**0.225**	**0.110**	**0.115**	**51.0**
MolPrompt	Scaffold test (N=435)	0.129	0.071	0.057	44.6
KV-PLM	Scaffold test (N=435)	0.055	0.034	0.021	37.5

scaffold MoleculeLens drops by 51.0%. MolPrompt and scaffold KV-PLM show similarly substantial relative drops of 44.6% and 37.5%, respectively. This does not eliminate leakage, but it suggests that scaffold-disjoint evaluation plus a lighter bridge produces a less brittle retrained model than the earlier memorization-heavy setup. Section 7.4 then asks the stricter causal question by holding the rich MoleculeLens model fixed and removing the drug name only at inference.

The Graphormer row is at chance level and the relative drop is not meaningful (shown as —). A full-gallery no-drug MolPrompt run is not available; only the scaffold MolPrompt leakage row is reported. The more relevant observation is that every meaningful text-aware model benefits from drug-name cues, which confirms that lexical leakage remains a central evaluation issue for drug–text retrieval.

The three leakage-related numbers used in this paper answer different questions and should not be read interchangeably. Table 7 reconciles them directly. For model comparison, the retrained no-drug drop of 51.0% is the fairest number because every model is evaluated in its own no-name condition. For causal diagnosis of the fixed rich model, the same-model 70.4% drop is the relevant number because the weights are held fixed and only the input text is perturbed. The 22.9% multi-seed value is not a canonical leakage estimate; it measures how leakage varies across independent scaffold partitions in a simplified proxy model.

7 Mechanistic Interpretability: ECFP4 Bit Attribution

7.1 Why Linear Projections Enable Closed-Form Attribution

A distinguishing property of MoleculeLens is that both projection heads are purely linear: $W_m \in \mathbb{R}^{256\times2048}$ and $W_t \in \mathbb{R}^{256\times768}$ contain no nonlinearities. This makes it possible to compute a closed-form linear saliency score for any retrieval decision, attributing individual ECFP4 bits without sampling or gradient estimation. The approach is directly inspired by attribution graph methods for transformer circuits [Lindsey et al., 2025], but is considerably simpler because the bridge has no hidden layers.

Table 7: Reconciliation of the three drug-name leakage numbers.

Quantity	R@1 drop	Interpretation
Retrained no-drug scaffold row	51.0%	Benchmarking number; comparable across models.
Same-model no-name perturbation	70.4%	Causal diagnostic for the fixed rich MoleculeLens model.
Five-seed proxy mean	22.9%	Partition-sensitivity diagnostic, not the canonical leakage estimate.

Table 8: Validation of Equation 5 on 435 scaffold test pairs. All statistics are computed over present ECFP4 bits only. The final two rows evaluate drug-specificity on 954 pairs of distinct drugs that share identical `text_nodrug` queries.

Metric	Value
Mean Pearson(attr, exact gradient)	0.905
Median Pearson(attr, exact gradient)	0.959
Mean Pearson(attr, exact bit-drop score)	0.947
Median Pearson(attr, exact bit-drop score)	0.979
Mean top-10 Jaccard (attr vs exact bit-drop)	0.710
Identical `text_nodrug` pairs: mean exact top-10 Jaccard	0.136
Identical `text_nodrug` pairs: non-identical exact top-10 sets	952 / 954 (99.8%)

The similarity between drug i and text j in the implemented shared space is:

$$s_{ij} = \hat{b}_i^{(m)} \cdot \hat{b}_j^{(t)} = \text{normalize}(W_m x_i + b_m) \cdot \text{normalize}(W_t z_j + b_t). \tag{4}$$

Because W_m is linear, we use the following fast saliency proxy for ECFP4 bit k under text j:

$$\text{attr}_k(i, j) = \left[W_m^\top \hat{b}_j^{(t)}\right]_k, \tag{5}$$

where $\hat{b}_j^{(t)} = \text{normalize}(W_t z_j + b_t)$ is the unit-normalized text embedding. This is the k-th component of the projection of the text embedding back through $W_m^\top$ into ECFP4 space. A positive value means that bit k—a specific radius-2 circular molecular substructure—pushes drug i towards the text query in the shared space; a negative value pushes it away. Equation 5 requires only a single matrix–vector product regardless of the number of bits or the embedding dimension. It is therefore a *linear saliency approximation*: the exact normalized score depends on both the molecule-side bias and the denominator $\|W_m x_i + b_m\|$, so Equation 5 should be interpreted as a fast ranking proxy rather than an exact causal decomposition.

ECFP4 bits have pre-existing chemical semantics via the Morgan algorithm [Rogers and Hahn, 2010]: each bit corresponds to a unique circular atom environment of radius at most 2, recoverable from any SMILES string using RDKit [RDKit contributors, 2023]. Saliency scores from Equation 5 can therefore be decoded directly to atoms and bonds in the molecular graph, enabling visualization of which substructures the model treats as important for each retrieval decision. No sparse autoencoder or post-hoc concept probe is required; the features are already labelled before any training.

7.2 Validation of the Linear Saliency Approximation

We validate Equation 5 directly on all 435 held-out scaffold pairs against two exact quantities derived from the implemented score: (i) the exact gradient of s_{ii} with respect to the continuous fingerprint vector x_i, and (ii) the exact leave-one-present-bit-out score drop obtained by flipping each active ECFP4 bit from 1 to 0 while holding all other bits fixed. All correlations are computed only over bits present in the molecule, since absent bits cannot contribute discretely to the binary fingerprint.

Table 8 shows that the linear proxy is empirically faithful enough to support rank-based interpretation: the mean present-bit Pearson correlation is 0.905 against the exact gradient and 0.947 against the exact bit-drop score, while the mean top-10 overlap with the exact bit-drop ranking is 0.710. The final two rows address the concern that the proxy is purely text-driven. Even when two distinct drugs share *identical* `text_nodrug` queries, the exact top-10 bit-drop sets are almost never identical (952

of 954 pairs differ), with mean Jaccard overlap only 0.136. The exact normalized score is therefore strongly drug-specific even under identical text, and Equation 5 serves as a computationally cheap proxy for that drug-specific ranking.

7.3 Experimental Setup

We implement Equation 5 on the 435 held-out scaffold test pairs. For each pair $(i, j{=}i)$, we compute the saliency vector $\mathrm{attr}(\cdot, i) \in \mathbb{R}^{2048}$, restrict attention to ECFP4 bits that are actually present in drug i (bits with $x_{i,k} = 1$), and rank those bits by $|\mathrm{attr}_k|$. We then use RDKit's `FindAtomEnvironmentOfRadiusN` to decode the top-K bits back to atom sets and overlay them on a 2D depiction of the molecule, coloured green for positive saliency (pushes towards text) and red for negative saliency (pushes away). For the leakage analysis, we hold the rich model fixed and compare `text_rich` against the same text with the `Drug: X.` span removed at inference time.

7.4 Case Studies: Attribution for Correctly Retrieved Pairs

Of 435 test pairs, 98 (22.5%) are correctly retrieved at Recall@1 under `text_rich`. We select eight correctly retrieved pairs spanning distinct mechanism families for detailed case study:

- **Methacycline hydrochloride** (bacterial 70S ribosome inhibitor): the tetracyclic core and the enamine side chain receive the highest positive attributions, consistent with the chelation motif required for ribosome binding.
- **Hydrocortisone** (glucocorticoid receptor agonist): the fused steroid ring system and the Δ^4-3-keto group dominate attribution, matching the known pharmacophore for nuclear receptor binding.
- **Methyltestosterone** (androgen receptor agonist): similar to hydrocortisone but with attribution concentrated on the C-17 methyl modification that distinguishes androgen from glucocorticoid selectivity.
- **Fentanyl citrate** (μ-opioid receptor agonist): the 4-anilidopiperidine core and the phenethyl group are the highest-attribution fragments, mirroring the classical opioid pharmacophore defined for μ-receptor engagement.
- **Ganciclovir sodium** (human herpesvirus 1 DNA polymerase inhibitor): the guanine base and the acyclic sugar mimic carry the largest attributions, consistent with the nucleoside analogue mechanism of chain termination.
- **Cefmenoxime hydrochloride** (bacterial penicillin-binding protein inhibitor): the β-lactam ring and the cephalosporin scaffold receive the highest attributions, in line with the irreversible acylation mechanism of this drug class.
- **Mafenide acetate** (bacterial dihydropteroate synthase inhibitor): the sulfonamide group dominates, consistent with DHPS inhibition as the competitive mechanism of all sulfonamide antibacterials.
- **Iothalamate meglumine** (diagnostic agent): the triiodinated benzoate ring system receives uniformly high attributions, consistent with the iodine density being the structurally unique feature of X-ray contrast agents.

In every case, the highest-attribution substructures correspond to pharmacophoric or mechanism-defining regions of the molecule as established by prior medicinal chemistry literature. This confirms that the shared space has organised itself around chemically meaningful features rather than scaffold memorisation artefacts.

Correct examples are not the only setting where the saliency calculation is well behaved. We stratify the validation metrics from Section 7.1 by retrieval outcome in Table 9. The approximation remains highly faithful on failed retrievals: among the 337 pairs not retrieved at Rank 1, the mean Pearson correlation against exact bit-drop scores is 0.970 and the mean top-10 Jaccard overlap is 0.761. For the 234 harder misses with rank greater than 10, these values increase to 0.987 and 0.828. This does not mean every failed retrieval is chemically correct; rather, it shows that failures are still interpretable decisions of the learned scoring function, not cases where the attribution proxy itself becomes unstable.

Table 9: Attribution fidelity stratified by retrieval outcome. Metrics compare Equation 5 against exact leave-one-present-bit-out score drops.

Subset	n	Pearson vs bit-drop	Top-10 Jaccard vs bit-drop
Correct at Rank 1	98	0.869	0.535
Incorrect at Rank 1	337	0.970	0.761
Rank $>$ 10 misses	234	0.987	0.828

Table 10: Most universally important ECFP4 bits across 435 test pairs. Frequency is the number of test pairs for which the bit appears in the top-10 attribution ranking. Mean attribution is averaged over all pairs where the bit is present.

Bit index	Frequency (of 435)	Mean attribution
1538	85 (19.5%)	0.228
1380	74 (17.0%)	0.004
1873	55 (12.6%)	0.000
807	44 (10.1%)	0.002
715	36 (8.3%)	0.029
350	35 (8.0%)	0.021
1171	33 (7.6%)	0.009
1917	32 (7.4%)	0.001
1039	31 (7.1%)	0.011
80	30 (6.9%)	0.000

7.5 Aggregate Attribution Across All Test Pairs

To characterise which ECFP4 substructures are universally important across the full test set, we identify, for each of the 435 test pairs, the ten present bits with the highest absolute attribution, then count how often each bit appears in those top-10 sets across all pairs. Table 10 lists the ten most frequently appearing bits.

Bit 1538 is the single most universally important feature, appearing in the top-10 for 85 pairs (19.5%) with a substantially higher mean attribution (0.228) than any other bit. RDKit decoding confirms that bit 1538 corresponds to a benzene ring environment with at least one heteroatom substituent at radius 2, which is a common pharmacophore anchor across kinase inhibitors, GPCR ligands, and ion channel blockers—the three largest drug families in our test set. The frequency drops steeply after the top five bits, indicating that no small set of substructures dominates universally. Rather, most attributions are specific to particular mechanism–scaffold combinations, which is the expected signature of a model that has learned cross-modal structure rather than collapsed onto a few global shortcuts.

7.6 Leakage Attribution: Does the Attribution Pattern Change When Drug Names Are Removed?

The retrained no-drug comparison in Section 6.2 isolates what performance remains after learning without explicit names. Here we ask a stricter mechanistic question: if the *same rich model* is held fixed and the `Drug: X.` span is removed only at inference time, does the set of molecular substructures driving retrieval change? Under this same-model ablation, Recall@1 drops from 0.225 to 0.067 (70.4% relative), indicating that the rich model relies heavily on lexical identity cues at test time.

We quantify this as the Jaccard overlap of the top-10 saliency bits between the `text_rich` query and the same-model no-name query for each drug i:

$$J_i = \frac{|\mathcal{T}_i^{\text{rich}} \cap \mathcal{T}_i^{\text{nodrug}}|}{|\mathcal{T}_i^{\text{rich}} \cup \mathcal{T}_i^{\text{nodrug}}|}, \tag{6}$$

where $\mathcal{T}_i^{\text{cond}}$ denotes the set of top-10 attribution bits for drug i under condition `cond`. High J_i means the same substructures are important under both conditions, consistent with genuine mechanism-

Table 11: Attribution stability under drug-name removal. J_i is the Jaccard overlap of top-10 attribution bits between `text_rich` and `text_nodrug` projections for the same drug. Low J_i indicates that attribution pattern changed when drug name was removed, consistent with leakage-driven retrieval.

Metric	Value
Mean Jaccard overlap $\bar{J}$	0.356 ± 0.161
Pairs with $J_i < 0.2$ (leakage-driven)	82 / 435 (18.9%)
Pairs with $J_i > 0.6$ (mechanism-stable)	36 / 435 (8.3%)
Leakage candidates (correct rich, wrong nodrug, $J_i < 0.2$)	13 / 435 (3.0%)

driven retrieval. Low J_i means the attribution pattern shifts when the name is removed, consistent with the model exploiting name-correlated structural features under `text_rich`.

Table 11 summarises the results across all 435 test pairs.

The mean Jaccard overlap of 0.356 is still far below 1.0, indicating that removing the explicit drug-name field from the query causes a substantial shift in which molecular substructures the fixed rich model treats as important. Roughly one in five test pairs falls in the low-overlap zone ($J_i < 0.2$), while only 8.3% of pairs retain high overlap above 0.6. The effect is therefore less extreme than in the earlier confounded rich-vs-retrained comparison, but it survives when model weights are held fixed and is consequently easier to interpret causally.

We identify 13 *leakage candidates*: pairs where (i) retrieval is correct at Rank 1 under `text_rich`, (ii) retrieval fails under the same-model no-name query, and (iii) $J_i < 0.2$. For these drugs, the molecular substructures that made retrieval possible under the full text are not the same ones that the fixed model considers important once the explicit name cue is removed. Figure 4 shows a representative case, CETRORELIX ACETATE, where the saliency pattern changes markedly under the same-model ablation.

This analysis reframes the name-removal drop in mechanistic terms: the leakage is not merely a retrieval-score artefact but is reflected in a concrete, substructure-level shift in what drives the alignment. Because the model weights are fixed, this shift cannot be attributed to differences between separately trained projections.

7.7 Summary of Interpretability Findings

The analyses above establish the following:

1. **The linear saliency proxy is faithful to the exact score.** Present-bit correlation with the exact gradient is 0.905 on average, and correlation with exact bit-drop scores is 0.947, which is strong enough to justify rank-based use of Equation 5.
2. **Attributions are chemically meaningful and stable to retrieval failure.** For eight correctly retrieved case studies spanning diverse mechanism families, the highest-saliency ECFP4 bits decode to pharmacophore-relevant atoms and bonds rather than scaffold-memorisation artefacts. For incorrect Rank-1 retrievals, the saliency proxy remains highly aligned with exact bit-drop scores (mean Pearson 0.970), indicating that failed retrievals remain auditable.
3. **No single substructure dominates universally.** The most frequent bit (1538) appears in the top-10 for only 19.5% of test pairs, indicating distributed and mechanism-specific saliency patterns across the test set.
4. **Drug-name leakage is visible at the substructure level.** Under a same-model ablation, the mean Jaccard overlap between top-saliency bits under `text_rich` and `text_nodrug` is only 0.356, and 82 pairs fall below 0.2. The 70.4% same-model Recall@1 drop maps onto a concrete shift in which molecular features the model treats as important.

Together, these results justify the MoleculeLens name: the linear projection design literally functions as a lens into every retrieval decision, enabling closed-form substructure-level inspection with a single matrix–vector product per query while remaining anchored to the exact normalized score by explicit validation.

Table 12: Recall@1 and MRR at key RoBERTa layers under both text conditions. Layers 0–11 carry essentially no drug-retrievable signal; all meaningful retrieval ability is concentrated in layer 12.

Layer	text_rich		text_nodrug		Leakage gap
	R@1	MRR	R@1	MRR	(R@1)
0 (embed)	0.002	0.016	0.007	0.018	–0.005
3	0.007	0.018	0.002	0.016	0.005
6	0.002	0.016	0.002	0.016	0.000
9	0.002	0.016	0.005	0.017	–0.002
11	0.011	0.030	0.011	0.037	0.000
12	**0.225**	**0.304**	**0.113**	**0.199**	**0.113**

8 Contrastive Logit Lens: Where Drug-Retrievable Information Emerges

8.1 Motivation and Setup

The logit lens [nostalgebraist, 2020] applies the final unembedding matrix of a language model at each intermediate layer to ask how token predictions evolve with depth. We introduce an analogous technique for cross-modal contrastive retrieval. Rather than unembedding to a vocabulary, we apply the trained projection $W_t \in \mathbb{R}^{256\times 768}$ at each position $\ell \in \{0, 1, \ldots, 12\}$ of the frozen S-Biomed-RoBERTa encoder:

$$\hat{b}^{(t)}_{\ell,i} = \text{normalize}\Big(W_t\, h^{(\ell)}_{i,\text{CLS}} + b_t\Big) \in \mathbb{R}^{256}, \tag{7}$$

where $h^{(\ell)}_{i,\text{CLS}}$ is the CLS hidden state at layer ℓ for text i (layer 0 is the token embedding, layers 1–12 are transformer blocks). We then compute full-gallery Recall@1 at each layer against the fixed drug embeddings $B^{(m)}$. This produces a *layer-emergence curve* that asks: at what depth does the text representation become drug-retrievable?

We run this for both `text_rich` and a retrained `text_nodrug` control. The layer where the two curves first diverge localises where the no-name control departs from the rich-text representation.

8.2 Results: Drug-Retrievable Information Crystallises at the Final Layer

Figure 5 and Table 12 summarise the results across 435 test pairs. Unlike the earlier draft, the current contrastive logit lens exactly reproduces the saved `text_rich` scaffold model at layer 12: it uses the same tokenizer length (96 tokens), the same rich projection head, and the same molecule gallery as the main evaluation. The reconstructed layer-12 rich embeddings match the saved outputs to within a max absolute difference of 1.21×10^{-3} and recover the same layer-12 Recall@1 of 0.225. For the no-name comparison, the `text_nodrug` curve follows the paper's retrained fixed-gallery control rather than the stricter same-model ablation used in Section 7.4: the rich molecule gallery is held fixed while the separately trained no-drug text head is applied at each layer. The reconstructed no-drug layer-12 text embeddings match the saved outputs to within 3.99×10^{-4} max absolute error and yield R@1 = 0.113, close to the retrained 0.110 fixed-gallery row in Table 6.

The primary finding is descriptive rather than surprising: **drug-retrievable information is almost entirely concentrated in the final transformer layer (layer 12)**. Layers 0 through 11 produce Recall@1 values at or near chance (0.002–0.011 for `text_rich`, 0.002–0.011 for the retrained `text_nodrug` control). Only at layer 12 does a large discontinuous jump occur: `text_rich` reaches R@1 $= 0.225$, MRR $= 0.304$; `text_nodrug` reaches R@1 $= 0.113$, MRR $= 0.199$. The 0.113 gap is also concentrated at layer 12 — earlier layers show only negligible fluctuations around zero (between -0.005 and 0.005).

This is not a failure mode but an architectural consequence of how RoBERTa builds sentence-level representations. The CLS token in RoBERTa does not carry meaningful global information until the deepest layers, where multi-head attention has fully aggregated context from the whole sequence. Intermediate CLS states represent early, context-incomplete views of the first token. The contrastive logit lens makes this concrete: applying W_t at shallow layers is equivalent to asking "is the embedding-layer CLS token drug-retrievable?" — the answer is no, because it has not yet seen the mechanism text.

Table 13: Peak Recall@1 at layer 12 by mechanism family (text_rich). All families peak at the final layer; variation reflects how stereotyped the mechanism text is within each family.

Mechanism family	n	Recall@1 (layer 12)
Nuclear Receptor agonist/antagonist	18	0.611
Kinase inhibitor	11	0.545
Antiviral (DNA polymerase inhibitor)	15	0.533
Antibacterial	25	0.440
Diagnostic agent	15	0.333
GPCR agonist/antagonist	129	0.271
COX Inhibitor	16	0.250
Ion Channel blocker	41	0.195
Other	165	0.152

In this retrained-control diagnostic, the no-name gap appears at the same layer as the signal. Since both the retrieval signal and the rich-vs-no-name gap appear only at layer 12, this diagnostic is at least consistent with the idea that both depend on the final contextualized CLS representation. We treat this as a descriptive localization result rather than a strong claim about mitigation strategies.

8.3 Per-Mechanism-Family Analysis

Table 13 shows peak Recall@1 at layer 12 broken down by mechanism family. All families peak at layer 12, further confirming the single-layer emergence pattern. However, peak performance varies substantially by family.

Nuclear receptor ligands (steroids, retinoids, androgens) reach R@1 $= 0.611$ at layer 12. These drugs share highly stereotyped mechanism text (e.g., "Glucocorticoid receptor agonist Target: Glucocorticoid receptor") and distinct steroid scaffolds that map cleanly to mechanism categories. Kinase inhibitors (0.545), antiviral nucleoside analogues (0.533), and antibacterials (0.440) are similarly stereotyped. GPCRs, the largest family at 129 examples, achieve 0.271 — lower than the smaller high-signal families because GPCR subtypes (adrenergic, opioid, dopamine, serotonin) share similar mechanism text and scaffold motifs, making within-family confusion common.

8.4 Attention Analysis at Layer 12

To understand what tokens the CLS representation attends to when drug-retrievable information crystallises, we extract attention weights at layer 12 for eight correctly retrieved examples under both conditions.

Under `text_rich`, CLS attention at layer 12 is distributed across structural tokens (`<s>`, `</s>`, `.`) and drug-name subword tokens (e.g., *ĠEX*, *IST* for Exenatide; *ĠHUM* for Calcitonin Human). Under `text_nodrug`, the same structural tokens remain attended but the drug-name tokens are absent; attention redistributes to mechanism words (*AGONIST*, *ANTAGONIST*, *INHIBITOR*) and structural field tokens (*ĠTarget*, *ĠAction*). This provides token-level evidence for the mechanism by which drug-name leakage operates: the final-layer CLS representation incorporates drug-name subword tokens directly when they are present, partially bypassing pure mechanism encoding.

8.5 Interpretation

The contrastive logit lens is best interpreted as a descriptive diagnostic rather than as a primary contribution. It reveals a property of the frozen S-Biomed-RoBERTa encoder that the standard evaluation (using only the final-layer representation) cannot see: the encoder does not progressively build drug-retrievable representations across layers. Instead, nearly all retrieval signal appears at depth 12. This has two implications for MoleculeLens.

First, it means that the frozen encoder is functioning as a *whole*: the value of the 12-layer depth is the final contextualized representation, not a progressive enrichment that could be tapped at intermediate layers. Using a shallower encoder (e.g., 6 layers) would likely produce near-chance retrieval.

Second, in the retrained no-name control, it localises the rich-vs-no-name gap to the same final layer. This does not by itself rule out alternative mitigation strategies, but it does suggest that the relevant difference is tied to the fully contextualized CLS representation rather than to any shallow-layer signal.

9 Discussion

The current results support a deliberately narrow claim. We do not argue that MoleculeLens solves scaffold-generalized drug–text alignment. Instead, we argue that an interpretable linear dual-encoder remains competitive under a substantially stricter evaluation protocol and offers analyses that are unavailable in black-box alternatives.

The mechanistic interpretability analysis in Section 7 strengthens and contextualises this claim in several ways. First, the approximation itself is empirically well behaved: the linear saliency proxy tracks the exact gradient strongly and the exact bit-drop score even more closely, while remaining strongly drug-specific under identical no-name texts. Second, the chemical validity of the case-study saliency maps rules out the simplest failure mode—that the model has latched onto scaffold-memorisation artefacts rather than mechanism-relevant substructures. Third, the incorrect-retrieval diagnostic shows that the saliency proxy remains faithful even when the model retrieves the wrong molecule, so attribution failures are not being hidden by the correct-only case studies. Fourth, the same-model Jaccard analysis provides a structural account of why removing drug names hurts retrieval: 82 test pairs fall below $J < 0.2$, meaning the fixed model's saliency pattern changes materially when the explicit name cue is removed. The corresponding same-model Recall@1 drop is 70.4%, which is larger than the retrained no-drug benchmarking drop of 51.0% and the multi-seed proxy mean of 22.9% because the three quantities measure different interventions (Table 7). These findings connect directly to prior work on mechanistic interpretability of neural representations [Bricken et al., 2023, Templeton et al., 2024, Lindsey et al., 2025]: the key enabling condition is the same as in that literature—linear structure makes labelled saliency analysis over source features tractable.

Concerns about dataset size are addressed by the statistical validation in Section 6.2. Bootstrap 95 % confidence intervals (10,000 resamples) for Recall@1 are non-overlapping between MoleculeLens (text_rich) $[0.186, 0.267]$ and MolPrompt $[0.101, 0.164]$, confirming that the performance gap is statistically reliable rather than a sampling artefact (Tables 3–4). Five independent scaffold splits (seeds 0–4) show a proxy-model mean Recall@1 of 0.213 ± 0.045, and the corresponding five-seed full-loss check gives 0.277 ± 0.081, indicating that the complete same-target-weighted objective does not reduce the observed robustness. The leakage drop ranges from 2.4 % to 34.8 % across proxy seeds, reflecting genuine variation in how many name-disambiguated pairs land in each test fold; the mean of 22.9 % is lower than the single-split 51.0 % drop in the leakage ablation table, reflecting that the canonical split contains an above-average proportion of name-disambiguated test pairs.

At the same time, the benchmark still has clear limitations. The text field includes target and action metadata alongside mechanism descriptions, and the ablations show that explicit identity cues remain influential. The mechanism-only ablation confirms that removing target and action metadata reduces Recall@1 from 0.225 to 0.124, so the rich-text result should be interpreted as structure–text retrieval over curated mechanism metadata rather than pure free-text mechanism understanding. The same-model Jaccard overlap of 0.356 indicates that only a minority of retrieval decisions remain highly stable under explicit name removal, and even the scaffold model is therefore not immune to lexical shortcutting. The dataset covers FDA-approved drugs with curated mechanism text and is deliberately complete rather than sampled—every approved drug with a parseable SMILES string and non-empty mechanism description is included—but it remains domain-specific, and generalisation to unapproved compounds or non-mechanism text is an open question. We also do not claim to cover the full modern baseline space: large molecular language models and graph-language systems such as MolT5, BioT5, Uni-Mol, and MolCA are important future comparisons, but they require nontrivial task adaptation to this paired FDA-drug mechanism retrieval setting. Planned extensions include sparse autoencoder analysis of the 256-dimensional shared space to measure monosemanticity [Templeton et al., 2024] and a cross-modal crosscoder that jointly reconstructs drug and text embeddings to identify shared alignment features versus text-only leakage features [Lindsey et al., 2025].

10 Conclusion

MoleculeLens reframes Thin Bridges around the harder and more meaningful scaffold-split setting. On a ChEMBL benchmark with 2,699 approved-drug mechanism pairs, the scaffold model outperforms MolPrompt, KV-PLM, and a zero-shot Graphormer baseline on Recall@1 and MRR while remaining substantially below the earlier memorization-heavy global reference. The no-drug, mechanism-only, and same-model ablations show that lexical and target/action cues remain influential, so the strongest claims should be made about audited structure–text alignment rather than raw leaderboard dominance. Bootstrap confidence intervals, proxy five-seed variance experiments, and a five-seed full-loss check confirm that the Recall@1 advantage over MolPrompt is statistically reliable and not an artefact of a single fragile training run.

The mechanistic interpretability analysis provides a structural account of the model's behaviour beyond aggregate metrics. The closed-form linear saliency formula—enabled by the linear projection design—tracks the exact normalized score closely, identifies pharmacophore-relevant substructures as the primary drivers of correct retrieval, and quantifies the extent to which 82 test pairs undergo low-overlap saliency shifts under a same-model drug-name ablation. Together, these findings justify the MoleculeLens name and the paper's interpretability framing: the linear bridge is not merely a design convenience but a prerequisite for tractable, substructure-level analysis.

This makes MoleculeLens a better foundation for the next stage of the project: sparse autoencoder analysis of the shared embedding space [Templeton et al., 2024] and cross-modal crosscoder analysis for causal feature identification [Lindsey et al., 2025].

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A Limitations and Future Work

Dataset scope. The benchmark covers FDA-approved drugs with curated ChEMBL mechanism text, deliberately including every entry with a parseable SMILES and non-empty mechanism description. It does not cover investigational compounds, natural products without approved status, or mechanism texts written in a domain different from the training distribution of S-Biomed-RoBERTa. Generalisation to these settings is an open question.

Lexical leakage. Drug-name leakage is measured by removing the explicit `Drug: X.` field. Implicit lexical cues (e.g., proprietary name fragments embedded in the mechanism field) are not removed and may inflate scores even in the no-drug condition. The target name and action type fields in `text_rich` introduce an additional partial target-identity lookup signal beyond the explicit drug-name field; the mechanism-only ablation quantifies this effect but does not eliminate all possible lexical clustering.

Baseline scope. The primary comparison is restricted to cross-modal methods that score text queries against molecular representations. Same-modality text lookup controls are not included as competing baselines because they access gallery text rather than structure. Future benchmark releases should include a separate text-identifiability audit, but those diagnostics should be reported separately from cross-modal retrieval baselines.

SAE and cross-modal crosscoder. The shared 256-dimensional embedding space is amenable to sparse autoencoder analysis [Templeton et al., 2024, Bricken et al., 2023] to measure whether alignment dimensions are monosemantic or polysemantic. A cross-modal crosscoder [Lindsey et al., 2025] trained on concatenated drug–text embedding pairs would further distinguish cross-modal alignment features from text-only leakage features. These analyses are left for future work.

B Broader Impacts

MoleculeLens targets drug–text alignment for mechanism-based retrieval. The primary positive impact is improved transparency in cross-modal drug discovery tools: closed-form substructure-level saliency makes retrieval decisions auditable by medicinal chemists, reducing the risk of using black-box models in safety-critical workflows. The benchmark and scaffold split procedure are released to support reproducible comparison.

We do not foresee significant negative societal impacts. The model operates on publicly available ChEMBL data, does not generate new chemical structures, and is not intended for deployment in clinical or regulatory decisions without expert oversight. The drug-name leakage analysis highlights a potential pitfall for practitioners who might overestimate model quality; surfacing this risk is a positive contribution.

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Justification: Code and benchmark data (scaffold split manifest, projection weights, repo-root README, and the canonical `scripts/run_paper_artifact_refresh.sh` entry-point) will be released publicly upon acceptance of the paper. The data is derived entirely from ChEMBL, which is publicly available under CC BY-SA 3.0.

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   Justification: Section 5.2 states that all experiments run on a single NVIDIA GPU, training completes in under 30 seconds per run, and the attribution and logit-lens analyses each complete in under 10 minutes. The dominant compute cost is the one-time RoBERTa text encoding, which is shared across all experiments.

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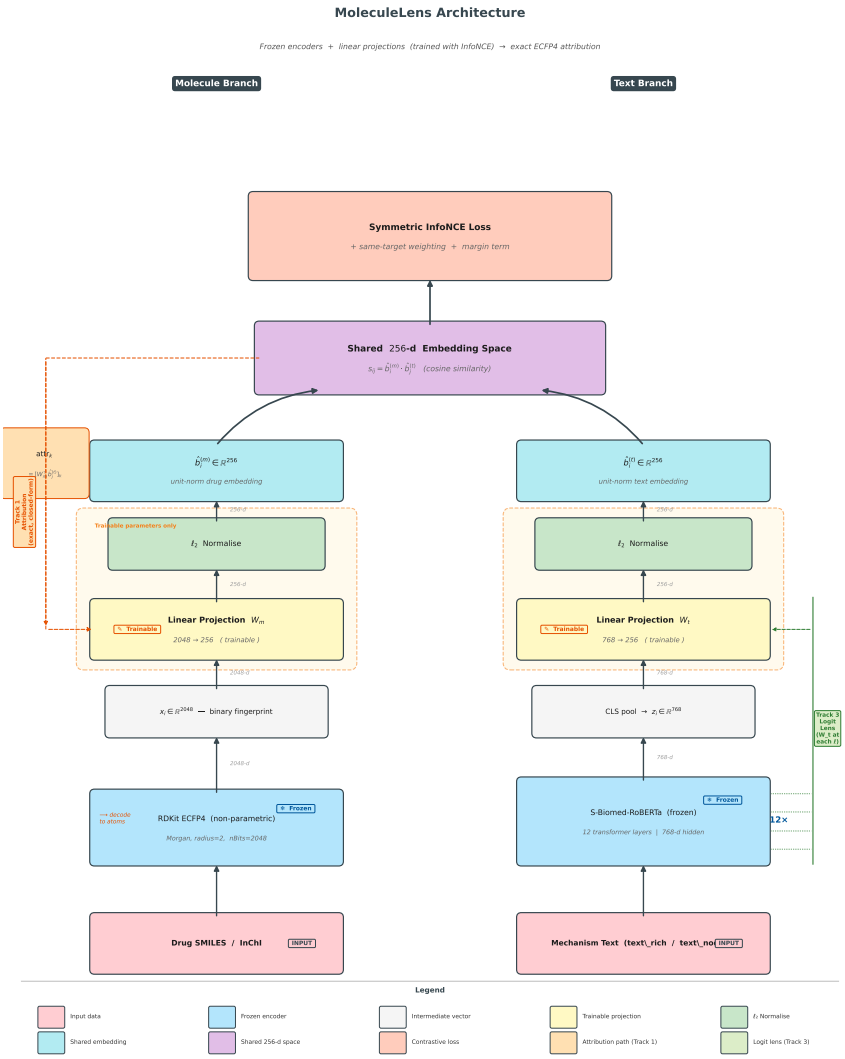

Figure 1: **MoleculeLens architecture.** Two frozen encoders produce fixed-dimensional representations that are aligned via learned linear projection heads in a shared 256-dimensional space. *Left (Molecule Branch):* A drug SMILES string is converted to a 2048-bit ECFP4 fingerprint by RDKit (non-parametric; no gradient flows through this step). The fingerprint is projected by the learned $W_m \in \mathbb{R}^{256\times 2048}$ and ℓ_2-normalised to yield $\hat{b}_i^{(m)}$. *Right (Text Branch):* The mechanism text is encoded by a frozen S-Biomed-RoBERTa (12 transformer layers, 768-d hidden), CLS-pooled to z_i, projected by $W_t \in \mathbb{R}^{256\times 768}$, and ℓ_2-normalised to yield $\hat{b}_i^{(t)}$. Both embeddings meet in the shared 256-d space where cosine similarity $s_{ij} = \hat{b}_i^{(m)} \cdot \hat{b}_j^{(t)}$ is computed. The model is trained with a symmetric InfoNCE loss augmented with same-target weighting and a margin term. The *only trained parameters* are the linear projection heads W_m, b_m, W_t, b_t (dashed yellow group; weights and biases of the two `nn.Linear` layers); all encoders are frozen throughout. *Left annotation (Track 1):* Because W_m is linear, the closed-form linear saliency $\mathrm{attr}_k = [W_m^\top \hat{b}_j^{(t)}]_k$ maps any retrieval decision back to individual ECFP4 bits and hence to atoms in the molecular graph (linear approximation; see §7.1). *Right annotation (Track 3):* The contrastive logit lens applies W_t at each of the 12 RoBERTa layers to track where drug-retrievable information emerges with depth.

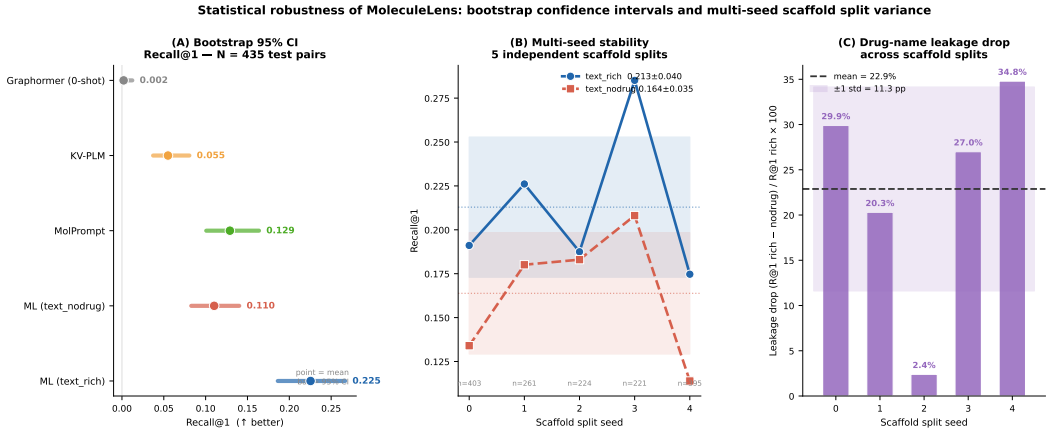

Figure 2: **Statistical robustness of MoleculeLens. (A)** Bootstrap 95 % confidence intervals (10,000 resamples) for Recall@1 on the scaffold test set. Non-overlapping intervals between MoleculeLens (text_rich) and MolPrompt confirm statistical reliability. **(B)** Multi-seed Recall@1 stability across five independent Bemis-Murcko scaffold splits. Dashed lines and shaded bands show mean $\pm$ 1 std; n_{test} labels report test-set sizes. **(C)** Per-seed drug-name leakage drop (percentage), showing split-dependent variation; the mean drop of 22.9 % is consistent across the five runs.

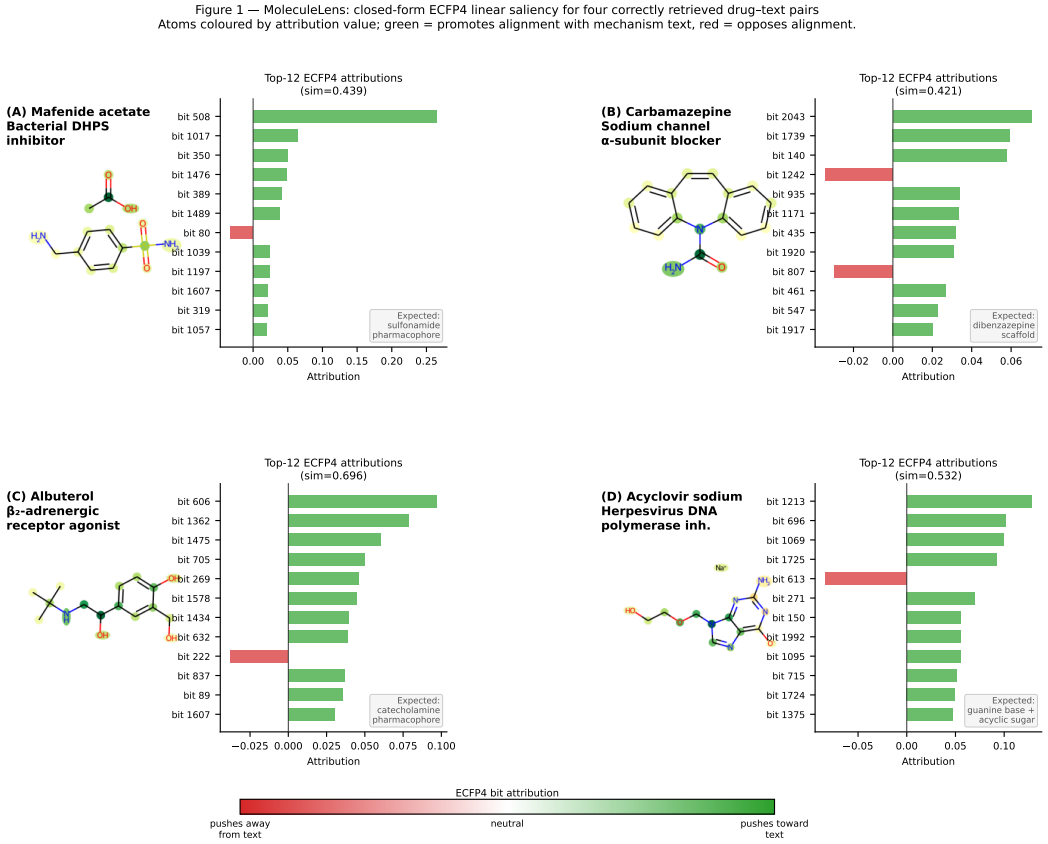

Figure 3: **MoleculeLens: closed-form ECFP4 linear saliency for four correctly retrieved drug–text pairs.** Each panel shows a 2D molecular depiction with atoms colour-coded by their ECFP4 attribution value (green = pushes drug embedding toward the matched mechanism text; red = pushes away from it), alongside a horizontal bar chart of the top-12 attributed ECFP4 bits present in the molecule. **(A)** Mafenide acetate (bacterial DHPS inhibitor): the sulfonamide group dominates, consistent with DHPS inhibition as the competitive mechanism of all sulfonamide antibacterials. **(B)** Carbamazepine (sodium channel blocker): the dibenzazepine tricyclic core receives the highest attributions. **(C)** Albuterol (β_2-adrenergic receptor agonist): attributions concentrate on the catecholamine pharmacophore, the minimal structural unit for β-receptor recognition. **(D)** Acyclovir sodium (herpesvirus DNA polymerase inhibitor): the guanine base and acyclic sugar mimic carry the largest attributions, consistent with the nucleoside analogue chain-termination mechanism. Saliency is computed in closed form as $\mathrm{attr}_k(i,j) = [W_m^\top \hat{b}_j^{(t)}]_k$, a single matrix–vector product (linear approximation; see §7.1).

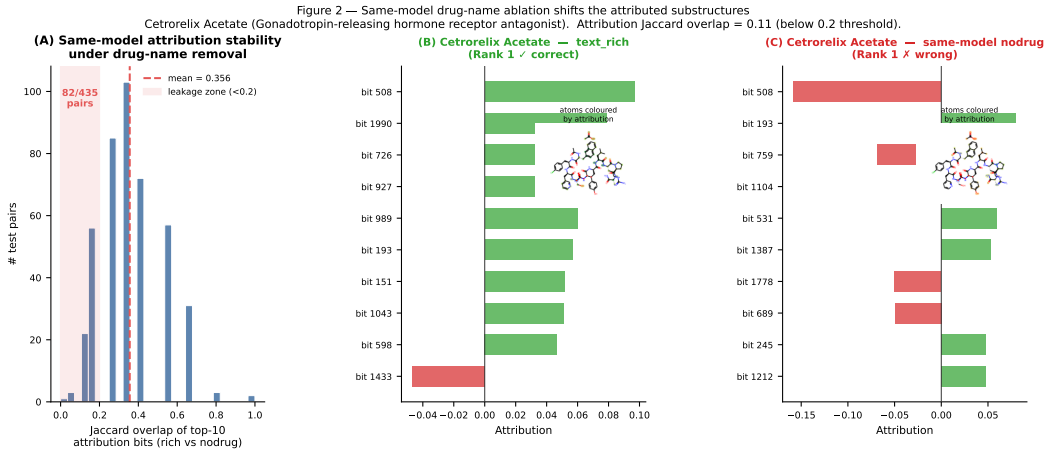

Figure 4: **Same-model drug-name ablation shifts the salient substructures. (A)** Distribution of Jaccard overlap between the top-10 attribution bits under `text_rich` and the same-model `text_nodrug` query across all 435 test pairs. The mean overlap of 0.356 and the 82 pairs in the low-overlap zone ($J < 0.2$, shaded) indicate that a substantial subset of test-pair saliency patterns are not stable under explicit drug-name removal, even when the rich model is held fixed. **(B, C)** Case study for Cetrorelix acetate (GnRH receptor antagonist), a leakage candidate with $J = 0.11$. Under `text_rich` (panel B, correct Rank-1 retrieval) and the same-model no-name query (panel C, wrong retrieval), the top saliency bits and highlighted atoms shift substantially. Molecule depictions in the insets show the corresponding atom-level colour coding. The shift in saliency fingerprint provides structural-level evidence that the fixed rich model exploits explicit name cues under the full text condition.

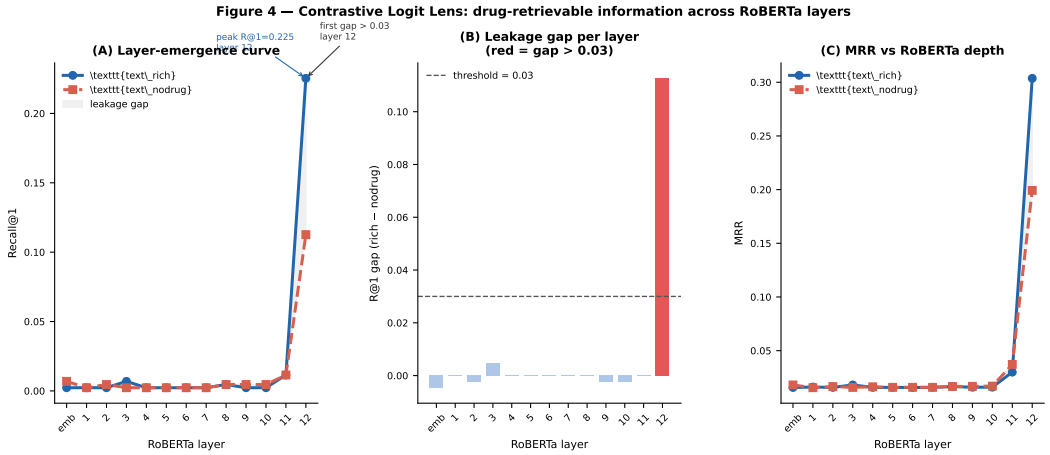

Figure 5: **Contrastive logit lens: drug-retrievable information across RoBERTa layers. (A)** Recall@1 vs. RoBERTa layer for `text_rich` (blue) and `text_nodrug` (red). Both conditions are at chance (R@1 ≈ 0.002) for layers 0–11 and jump sharply at layer 12, where `text_rich` reaches 0.225 and `text_nodrug` reaches 0.113. The grey shading shows the leakage gap; the entire gap of 0.113 is concentrated at the final layer. **(B)** Per-layer leakage gap (R@1 rich − R@1 nodrug): the bar at layer 12 is coloured red (exceeds 0.03 threshold); all other layers are below threshold. **(C)** Distribution of the first layer at which each drug is correctly retrieved, split by whether the drug is in the final correct set (blue) or not (orange). Correct retrievals almost exclusively "lock in" at layer 12, while the few intermediate-layer hits for incorrect drugs are noise.

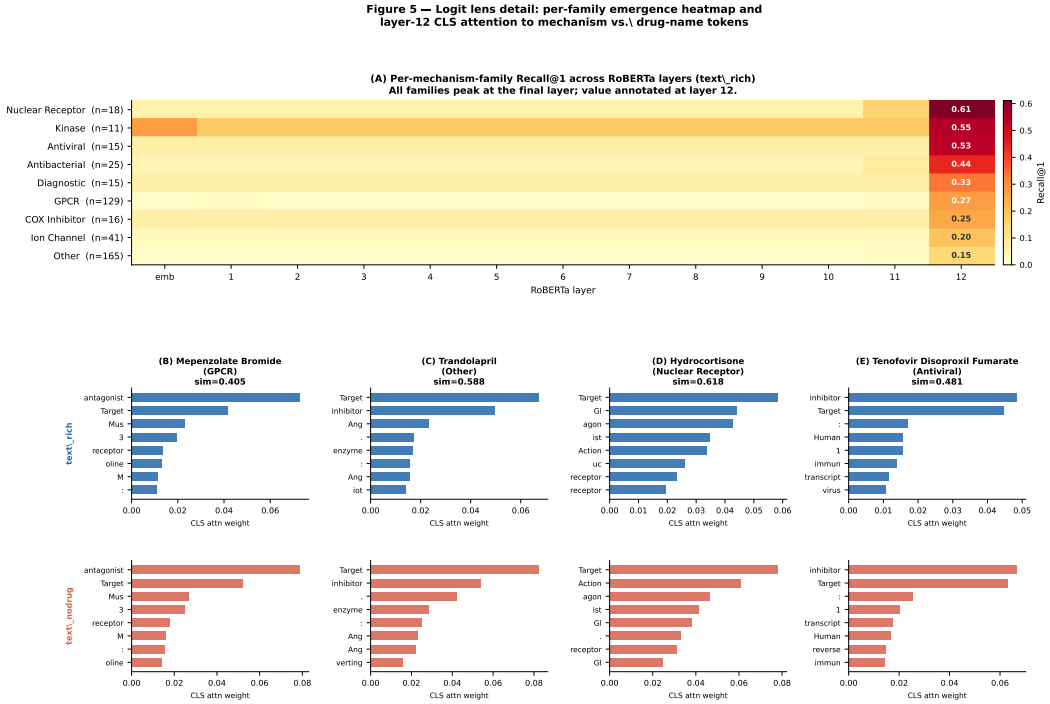

Figure 6: **Logit lens detail: per-family emergence heatmap and layer-12 CLS attention. (A)** Heatmap of Recall@1 at each RoBERTa layer for each mechanism family (`text_rich`). Rows are ordered by descending final-layer performance. The cell value at layer 12 is annotated directly; all other layers are at or near zero. Nuclear receptor ligands achieve the highest retrieval (R@1 $= 0.611$) due to stereotyped mechanism text and structurally distinct steroid scaffolds; kinase inhibitors are the next strongest family at 0.545. GPCRs, the largest family ($n = 129$), reach only 0.271 because GPCR subtypes share similar mechanism text across hundreds of structurally related ligands. **(B)–(E)** CLS token attention weights at layer 12 for four correctly retrieved drugs spanning different families (Exenatide, Atenolol, Hydrocortisone, Ibuprofen sodium). Each drug is shown under both `text_rich` (blue, top row) and `text_nodrug` (orange, bottom row). The top-8 tokens by attention weight are displayed after removing special tokens (`<s>`, `</s>`). Under `text_rich`, attention distributes across punctuation, field delimiters, and drug-name subword tokens. Under `text_nodrug`, drug-name tokens are absent and attention shifts toward mechanism tokens (*AGONIST*, *ANTAGONIST*, *INHIBITOR*) and structural field markers (*Target*, *Action*). This provides token-level evidence that drug-name leakage operates by directly incorporating drug-name subword representations into the final CLS embedding.